

Carlos Serrouya

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WORK EXPERIENCE AND PERSONAL PROJECTS

Data Science Intern *Alberta Biodiversity Monitoring Institute* 2023-2024

- Developed machine learning models for animal detection, snow depth, and snow coverage in camera trap images.
- Implemented U-Net neural network, transfer learning, multiple regression, and random forests models using Keras/TensorFlow and Scikit-learn.
- Optimized image processing with NumPy/Pillow, reducing RAM and VRAM usage during training.
- Created object contouring, detection, and outlier detection algorithms.
- Managed large SQL datasets on AWS, integrated multiple regression models into SQL functions.
- Presented machine learning results to a non-technical panel, achieving 84% snow depth accuracy and 96% coverage classification.
- Reduced false positives in Microsoft's Megadetector by 72%, false negatives by 9%.
- Visualized model performance data with Matplotlib and Seaborn.
- Processed bird audio recordings from ARUs (Autonomous Recording Units), automating classification using BirdNET.
- Used Python scripting to automate image/audio conversion on Linux servers, reducing file sizes for backup.

Personal Project *NFL Score Prediction Model* 2023

- Used BeautifulSoup to scrape data from sports stats websites.
- Developed a predictive model for NFL game outcomes with 67% accuracy on test data.

Server/Bartender/Host *Village Idiot Bar and Grill* 2016-2023

- Worked in a high-paced environment, handled customer interactions, and managed cash transactions.
- Operated as part of a team, successfully mitigated conflicts and handled thousands of dollars each shift.

Basketball Coach *Revelstoke Secondary School* 2019-2021

- Volunteered to coach U13 basketball team while serving as the varsity team captain.
- Organized drills, coached games, and led practices.
- Awarded the Terry Beitel award for community athletic leadership and achievement.

EDUCATION AND CERTIFICATIONS

University of Calgary		Coursera	
BSc Computer Science	2022-2026	Advanced Learning Algorithms	2023
BSc Mathematics	2021-2026	Regression & Classification	2023

ADVANCED SKILLS

Python	Statistical Analysis	Image Processing
AWS	Anomaly Detection	Machine Learning